FLAVOURBENCH: EXECUTABLE CULINARY REWARD MAPS FOR LANGUAGE MODEL EVALUATION AND POST-TRAINING

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ABSTRACT

Open-ended language-model evaluation often substitutes another model or a small preference panel for a missing answer key. We introduce FLAVOURBENCH, which instead compiles dense answer maps from a versioned culinary environment. Each task asks for a three-ingredient portfolio from eight candidates; before inference, EPICURE scores all 56 portfolios. We evaluate 27 frontier endpoints on the same 534 substitution, pairing, and constraint tasks, yielding 14418 complete model–task observations. Anchor-cluster bootstraps and multiplicity-controlled paired tests resolve 101 of 351 model contrasts. Grok 4.6 has the largest point estimate (65.1), but the corrected evidence does not identify a unique best endpoint. The ranking replicates across independently compiled panels and remains similar under alternative metrics, task filters, family weights, and three public EPICURE checkpoints. We then run a preregistered, three-seed post-training study. LoRA SFT of a pinned Qwen3-0.6B checkpoint on 270 EPICURE-optimal answers improves its score on 84 anchor-disjoint maps by 13.30 points over a format- and label-matched control (95% CI 6.52–20.29, $p = 1.70 \times 10^{-4}$), and the effect replicates on all 534 public maps. The replication gain is 11.73 points (95% CI 8.98–14.54). On the primary split, both trained arms parse every response while the format control does not improve on the base model. The release contains prompts, exhaustive reward maps, raw responses, training and evaluation manifests, statistical plans, code, and an offline verifier.

1 INTRODUCTION

An executable evaluator gives a benchmark a stable object to measure. Programs can be tested, database states inspected, and games scored. Many open-ended language tasks lack that structure: human votes are costly, model judges inherit model-dependent biases, and exact-match keys collapse plausible answers into a binary outcome (Zheng et al., 2023; Chiang et al., 2024). The result may be scalable without being auditable.

FLAVOURBENCH creates executable evaluation for culinary decisions. It starts from EPICURE, a versioned representation of 1,790 ingredients in a 300-dimensional space (Radzikowski & Chen, 2026b;a). A compiler turns substitution, pairing, and constrained-composition operations into three-of-eight selection problems. It then enumerates all $\binom{8}{3} = 56$ legal portfolios and records a continuous reward map before any evaluated model is called. Scoring is a deterministic lookup, not a post-hoc interpretation of generated text.

This construction does not make EPICURE "correct" in a universal culinary sense. The primary score measures agreement with one released reward function under one published task distribution. The benchmark is useful only if that estimand is explicit and its dependence on the reward map can be tested. We therefore combine a complete common-task design with panel replication, clustered inference, roster and metric sensitivity, rescoring under three public checkpoints, and an external substitution check.

Our contributions are:

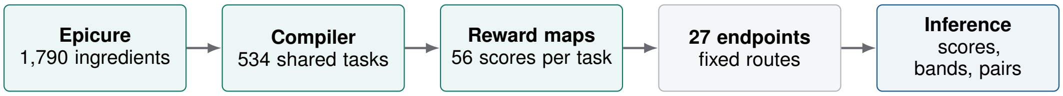

Figure 1: FLAVOURBENCH freezes the tasks, every candidate reward, and the inference contract before model execution. Evaluated models neither access EPICURE nor judge one another.

Table 1: Complete-core task families. Each contains 178 tasks and uses the same three-of-eight response contract.

Family	Decision	Environment utility
Substitution	Replace an anchor with a coherent three-item portfolio	0.8 anchor similarity +0.2 portfolio coherence
Pairing	Choose three ingredients to accompany an anchor	0.65 anchor affinity +0.35 portfolio coherence
Constraints	Choose a portfolio satisfying diet and processing limits	0.7 anchor similarity +0.3 coherence; invalid sets score zero

1. an executable benchmark that freezes 29904 graded portfolio rewards before model execution and exposes a single 0–100 score;
2. a rectangular 27-by-534 frontier evaluation with simultaneous uncertainty, all 351 paired contrasts, and score-blind common-core construction;
3. post-collection tests of panel, task-filter, score-definition, family-weight, and reward-map dependence, plus label-independent validation against observed substitutions; and
4. a preregistered three-seed study showing that EPICURE-optimal SFT transfers to unseen reward maps beyond answer-format learning, with a public-map replication and content-addressed training evidence.

2 EXECUTABLE BENCHMARK CONSTRUCTION

2.1 REFERENCE ENVIRONMENT AND TASK FAMILIES

The primary task maps bind one exact EPICURE data bundle and application build by content hash. Its original training seed and source revision were not recovered, so we treat it as a fixed reference function rather than independently established culinary ground truth. Changing the artifact creates a new benchmark identity. Section 5 evaluates three independently published, immutable sibling checkpoints.

For task t, let $\mathcal{A}_t$ be the 56 three-item subsets of its eight candidates and $u_t(a)$ the environment utility of portfolio a. Invalid portfolios under a dietary or processing constraint receive zero; valid utilities are normalized within task:

$$s_t(a) = 100\,\frac{u_t(a) - \min_{b\in\mathcal{A}_t} u_t(b)}{\max_{b\in\mathcal{A}_t} u_t(b) - \min_{b\in\mathcal{A}_t} u_t(b)}. \tag{1}$$

Each task has a unique 100-point optimum. Its exact chance value is the mean of its 56 frozen scores, not a generic percentage.

The ranked set contains 534 tasks, balanced across two independently compiled panels and three decision families (Table 1). The panels have disjoint task identifiers; tasks sharing an anchor ingredient form one cluster for inference. Candidate labels and order follow a task-derived hash. No evaluated model proposes candidates, distractors, or scores.

Prompts end with an exact `FINAL_SELECTION` marker and require three distinct labels. The parser reads the final marker line, sorts the labels as an unordered set, and performs one reward-map lookup; free-form explanation does not affect the score. Missing markers, duplicate labels, provider failures, and abnormal finishes remain in the execution ledger but cannot enter the shared core.

The score maps are graded rather than exact keys with 55 equivalent mistakes. Substitution and pairing tasks each have 56 distinct scores at the median, exact chance means of 45.2 and 45.0, and

Table 2: Deterministically selected common-core examples. Parentheses give the frozen task score; the supplement includes complete prompts, choices, score maps, and raw responses.

Family	Prompt	Higher-ranked selection	Lower-ranked selection
Substitution	Select three alternatives to ‘boursin cheese‘ that best preserve its dairy role, regional context, and mutual portfolio coherence.	caciocavallo, fromage blanc, grana padano (100)	fromage blanc, quail egg, goose egg (15)
Pairing	Select the three-ingredient bundle with the strongest learned pairing to ‘sweetbread‘ and the best internal coherence.	parsnip, cipollini onion, porcini mushroom (92)	parsnip, pickled onion, peppadew pepper (0)
Constraints	For a dish centered on ‘potato‘, select three vegetarian complements with NOVA processing level at most 2; among valid portfolios, maximize learned pairing and coherence.	parsley, black pepper, thyme (100)	cheese, bay leaf, parsley (0)

median top-two gaps of 5.6 points. Constraint maps are deliberately sparse: infeasible portfolios score zero, producing an exact chance mean of 3.4 and a median top-two gap of 37.3. The examples in Table 2 show why partial credit matters: non-optimal portfolios can be far apart even when both satisfy the response syntax.

2.2 Primary score

For model m, panel p, family f, and selected task set T_{pf}, let Y_{mt} be the frozen portfolio score. The primary metric is

$$F_m = \frac{1}{6} \sum_{p=1}^{2} \sum_{f=1}^{3} \frac{1}{|T_{pf}|} \sum_{t \in T_{pf}} Y_{mt}. \tag{2}$$

Because all six strata contain 89 tasks, this equals the ordinary mean over the 534 cells. Every ranked endpoint has one valid response for every selected task. Failures are not assigned a culinary score and are not dropped model by model; task eligibility is determined jointly from status and parser validity before any selected portfolio or score is loaded.

3 Evaluation and inference

3.1 Endpoint panel and common core

The panel covers 27 endpoints from 16 laboratories: OpenAI, Anthropic, Google, xAI, Meta, Kimi, Qwen, Z.ai, DeepSeek, ByteDance, Thinking Machines, MiniMax, NVIDIA, Mistral, Tencent, and Cohere. It includes multiple endpoints from several laboratories to separate lab-level branding from endpoint-level behavior. The manifest binds requested and returned identities, model version, provider route, supported controls, completion contract, and collection block. Automatic fallback is disabled. When transport validation required a route change, the complete model block was rerun; successful cells from superseded blocks were never pooled. Exact endpoints and routes are listed in Appendix B.

Each panel compiler creates 160 candidates per family. Within every panel–family stratum, the analysis selects the first 89 tasks eligible for all 27 endpoints under a fixed SHA-256 order. Selection code sees only response status and parser validity until task identifiers are fixed. The result is a rectangular 27-by-534 matrix rather than a leaderboard in which models answer different task subsets.

3.2 Uncertainty, multiplicity, and stability

The content-addressed analysis plan fixes all inferential choices. We draw 50000 bootstrap samples over the 534 anchor clusters, keeping tasks that share an anchor together, and report both pointwise score intervals and a simultaneous max-t band over all 27 scores. Every pair of endpoints is compared on the same 534 tasks using a two-sided anchor-cluster sign-flip test with 100000 Monte Carlo draws; Holm correction controls familywise error across all 351 comparisons. A separate Holm family

Table 3: Compact robustness summary. Rank statistics compare with the complete primary point order; these diagnostics do not redefine the confirmatory score.

Diagnostic	Result	Reading
Independent panels	$r = .89$, $\rho = .80$	broad replication
Crossed-design coefficient	$G = .936$	high relative-score reliability
Half-size subsets	median $\rho = .952$	stable order, unstable winner
Leave-one-endpoint task filter	$\rho = .955$	limited roster dependence
Alternative score definitions	$\rho \geq .938$	broad metric robustness
Moderate family weights	$\rho \geq .980$	little weight dependence
Public reward checkpoints	$\rho = .903$–$.957$	broad reward-map robustness

tests each endpoint against the exact taskwise random-portfolio mean. Bootstrap rank intervals and contiguous non-rejection groups accompany point ranks.

Two diagnostics ask whether 534 tasks are enough. First, family–panel-stratified subsets of 30, 60, 90, 150, and 270 tasks are drawn without replacement 5,000 times and compared with the full point order. Second, a crossed model-by-task variance decomposition estimates a descriptive generalizability coefficient for relative endpoint decisions. Neither calculation turns these fixed tasks or endpoints into random samples from an unspecified population.

We also rerun task selection after omitting each endpoint; compare selected with unselected compiler candidates; recompute ranks with chance-adjusted gain, action percentile, and exact-optimum rate; and enumerate all one-percentage-point family weights from 0.20 to 0.50. These are retrospective sensitivity analyses, not additional confirmatory rankings.

4 Results

4.1 Leaderboard and resolved comparisons

Grok 4.6 has the largest point estimate, 65.1, with simultaneous 95% interval [61.0, 69.2] (Figure 2). Of the 351 prespecified pairwise contrasts, 101 survive Holm correction. The upper ranks therefore form an unresolved group: the experiment supports many broad differences, but not a unique best endpoint. Every point in the figure is computed from the identical 534-task core.

4.2 Replication and design sensitivity

The two panel-specific score vectors correlate at Pearson $r = 0.89$ and rank $\rho = 0.80$. The crossed-design relative-decision generalizability coefficient is 0.936; the same variance model estimates 329 balanced tasks for 0.90. At 270 tasks, the median rank correlation with the full point order is 0.952 (empirical 95% range 0.897–0.980), yet the full point leader survives only 46.1% of draws. This distinction matters: the global score is stable while the identity of a single winner is not.

Omitting the endpoint that most constrained eligibility and reapplying the frozen task hash retains 71.0% of tasks; among the remaining models, rank correlation is 0.955 and the largest score shift is 1.37 points. None of 42 comparisons between selected and unselected task candidates survives Holm correction, and the largest standardized difference is 0.27. Alternative score definitions retain rank correlations of at least 0.938; across 696 moderate family-weight settings, the minimum is 0.980. The point leader alternates between Grok 4.6 and Gemini 3.1 Pro, again arguing against a definitive winner claim.

5 Reward-map and construct validity

5.1 Replacing the reward map

The primary runtime is reproducible by hash but lacks a recovered training lineage. We therefore rescore all 14418 fixed model decisions with Epicure-Cooc, Epicure-Core, and Epicure-Chem at immutable public revisions (Radzikowski & Chen, 2026a). This post-hoc analysis does not call

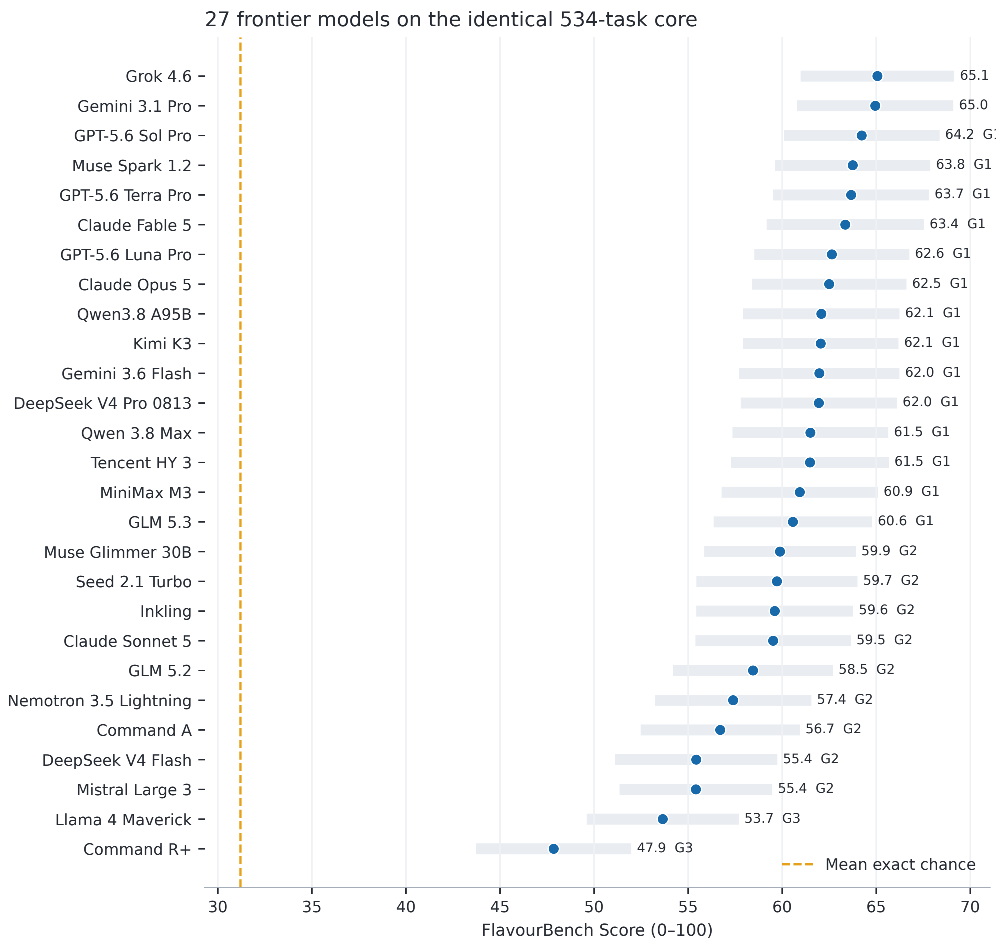

Figure 2: FlavourBench Score and simultaneous 95% max-t intervals. Group numbers summarize the multiplicity-controlled pairwise graph; point ranks inside a group are not established total orders. Exact values and rank intervals appear in Appendix A.

any endpoint again. Median within-task rank correlation between the primary and replacement 56-action maps is 0.660–0.752; exact optima agree on 27.7%–38.4% of tasks. Despite that local change, aggregate model-rank correlations are 0.903– 0.957 and 86.9%–91.7% of pair directions agree. The same sampled point leader appears under both maps in only 38.9%–53.8% of bootstrap draws, supporting broad order rather than a fixed winner.

5.2 Observed substitutions

Recipe1MSubs derives standardized substitutions from recipe-user comments (Fatemi et al., 2023). A hash-bound protocol, fixed before checkpoint scores were inspected, uses exact ingredient-token matches, deduplicates directed source–target pairs, and ranks each observed target by source–candidate cosine similarity. The primary candidate set is restricted to the target's existing food group so that broad category recognition cannot clear the null. Pair percentiles are averaged within source and then equally across sources; 50,000 source-cluster bootstrap draws form intervals, with Holm correction across three checkpoint tests.

Exact matching yields 1,469 unique directed test pairs over 357 sources (Table 4). All three intervals exclude the 0.5 chance percentile after Holm correction ($p_{\text{Holm}} < 10^{-4}$). The result persists on 594 pairs absent from Recipe1MSubs training. These labels are external to EPICURE fitting, but Recipe1MSubs and part of EPICURE share Recipe1M ancestry. The test is therefore label-independent

Figure 3: Top: independent-panel scores for all endpoints. Bottom: stability of score-blind task subsets relative to the complete 534-task point order. Panel agreement and subset stability support the global score; leader preservation shows why the top point estimate is not definitive.

Table 4: Observed-substitution validation. Percentile is the equal-source rank against same-food-group candidates; "novel" retains pairs absent from Recipe1MSubs training.

Checkpoint	Percentile [95% CI]	Novel pairs	Hit@10
Cooc	0.806 [0.788, 0.824]	0.754	0.133
Core	0.800 [0.781, 0.819]	0.735	0.172
Chem	0.780 [0.761, 0.798]	0.718	0.155

convergent validation, not corpus-independent validation, and it does not validate the unrecovered primary runtime directly.

6 Culinary profiles

The aggregate score does not collapse the three decision families into the same behavior (Figure 13). Gemini 3.1 Pro has the highest substitution mean, Kimi K3 the highest pairing mean, and Grok 4.6 the

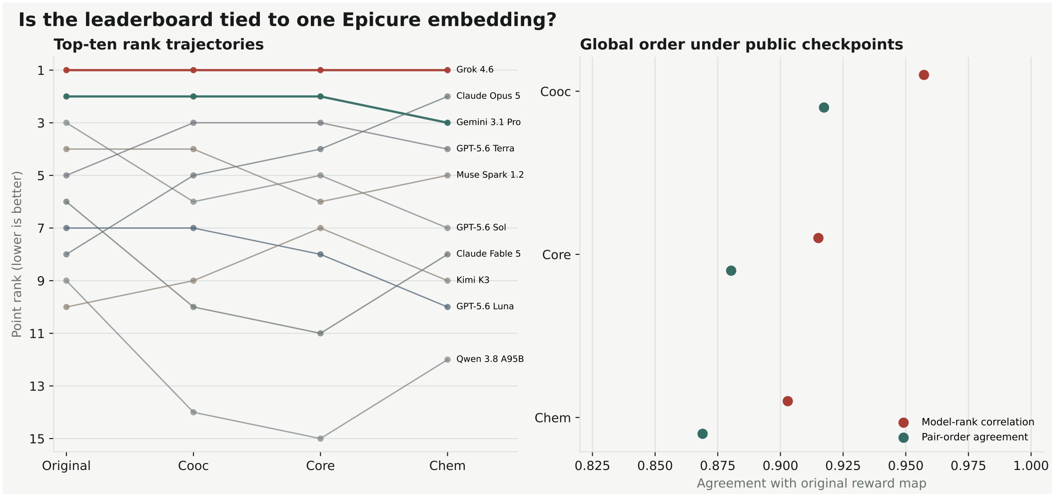

Figure 4: Reward-map sensitivity with prompts and model decisions held fixed. Left: rank trajectories for the primary top ten. Right: aggregate rank and pair-direction agreement with the primary map.

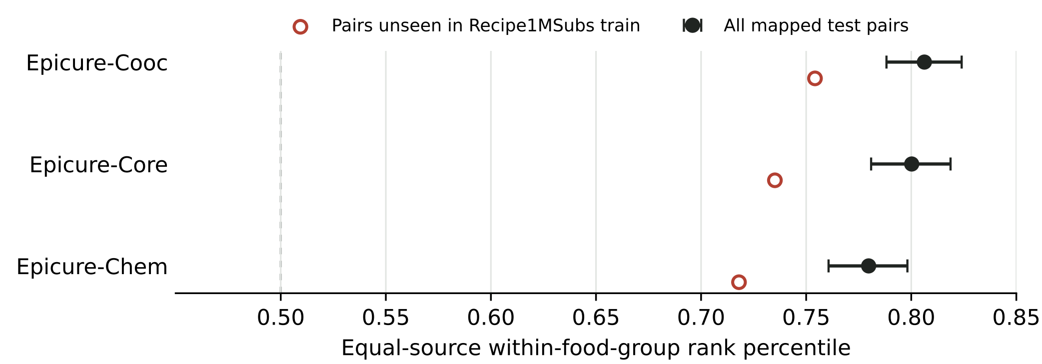

Figure 5: Observed targets ranked against same-food-group alternatives. Filled points use all mapped test pairs; open points retain only directed pairs absent from Recipe1MSubs training.

highest constraint mean; no endpoint leads every family. These values should be read within family rather than as a difficulty ordering because the frozen chance surfaces differ substantially. The decomposition is nevertheless useful diagnostically: a lab can distinguish a model that retrieves plausible substitutes from one that composes coherent bundles or reliably respects feasibility constraints.

7 Reward transfer beyond format learning

The exhaustive maps can supervise a policy as well as score one. We test the narrow claim that Epicure supervision transfers to unseen maps beyond learning the required response syntax. The protocol and analysis code were public before optimization outcomes were generated. It pins Qwen3-0.6B at one repository revision (Yang et al., 2025), partitions 270 training, 72 validation, and 84 primary-transfer tasks by disjoint ingredient anchor, and reserves the 534 leaderboard maps for a secondary replication. Candidate sets and prompt hashes are also disjoint across the lab splits.

Three seeds receive three epochs of all-linear LoRA SFT (Hu et al., 2022) with rank 16, effective batch size 16, learning rate 10^{-4}, completion-only loss, and the final checkpoint without validation-based selection. The treatment completion is each task's Epicure optimum. The control rotates those same A–H portfolios onto different prompts within family and panel, preserving every prompt, row count, completion length, and portfolio-label histogram while preventing accidental optima. Greedy

Table 5: Reward transfer. Scores average three training seeds except the unmodified base. Δ is Epicure SFT minus format control; the public-map row is a declared secondary replication opened only after the primary analysis was sealed.

Evaluation	Base	Format control	Epicure SFT	Δ [95% CI]	p
Anchor-disjoint transfer ($n = 84$)	29.99	30.90	44.20	+13.30 [6.52, 20.29]	1.70×10^{-4}
Public-map replication ($n = 534$)	28.56	31.60	43.33	+11.73 [8.98, 14.54]	1.00×10^{-5}

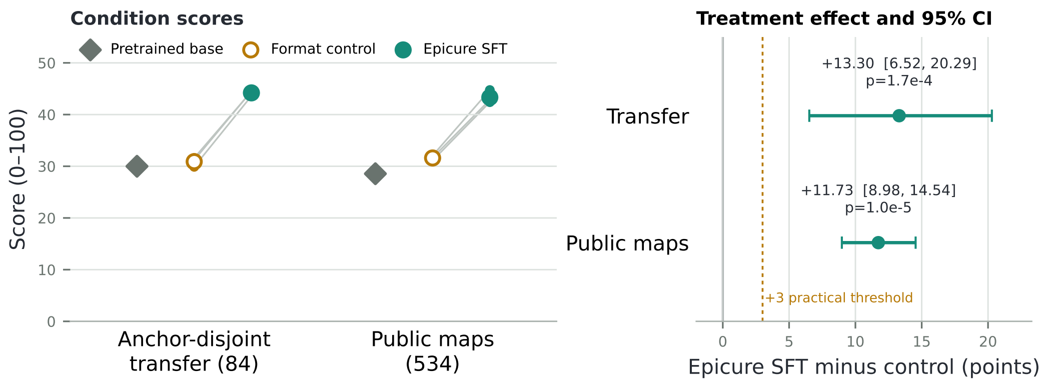

Figure 6: Held-out scores and matched treatment effects. Small points are training seeds; connecting lines pair the control and treatment initialized with the same seed. The right panel reports the prespecified crossed-bootstrap interval. The dashed line marks the three-point practical threshold, not a null hypothesis.

decoding and a 64-token limit are fixed for every arm. Unparseable completions remain in the score at zero.

The preregistered estimand is the equal-family treatment-minus-control score difference. A 50,000-draw crossed bootstrap resamples matched training seeds and anchors within the six family–panel strata; a two-sided 100,000-draw sign-flip test operates on anchor effects averaged over seeds. The protocol defines three points as a practically material gain. There is one primary contrast, so it receives no multiplicity adjustment.

On the 84 anchor-disjoint tasks, EPICURE SFT scores 44.20 versus 30.90 for the format control, a 13.30-point gain (95% CI 6.52–20.29, $p = 1.70 \times 10^{-4}$). All three primary seed effects are positive. Both trained arms parse 100.00% of responses; the base parses 89.29%. Format training alone changes the base score by only 0.90 points (95% CI -4.91–6.85, $p = 0.790$). The public-map replication yields a 11.73-point effect (95% CI 8.98–14.54, $p = 1.00 \times 10^{-5}$; Figure 6); all three replication seed effects are positive. Appendix H reports every seed and the descriptive family decomposition.

The control is empirically consequential. On the public maps it improves on the base by 3.05 points (95% CI 0.46–5.68, $p = 0.021$), consistent with removing parse failures. A treatment–base comparison would fold that syntax gain into the reward effect. Treatment–control differences are positive in all six evaluation-by-family cells and largest for constraints in both evaluations; these family rows are descriptive rather than separately tested.

These results isolate learning of this released reward function from answer-format learning. They do not establish human preference, cooked-food quality, general capability, reinforcement-learning improvement, or transfer outside the culinary environment.

8 RELATED WORK

General evaluation includes broad suites such as HELM (Liang et al., 2023), difficult knowledge tests such as MMLU-Pro and GPQA (Wang et al., 2024; Rein et al., 2024), contamination-resistant sets

(White et al., 2025; Jain et al., 2024), and human-preference arenas (Chiang et al., 2024). BetterBench emphasizes explicit benchmark documentation and validity claims (Reuel et al., 2024). Executable benchmarks replace preference inference with observable outcomes: SWE-bench runs repository tests (Jimenez et al., 2024), BFCL checks function calls (Patil et al., 2025), and τ-bench inspects database state (Yao et al., 2025). FLAVOURBENCH applies the same principle to a dense domain reward surface. RewardBench instead evaluates learned reward models on chosen–rejected pairs (Lambert et al., 2024). Our reward source is a versioned program rather than a learned preference model, and the transfer experiment tests whether its supervision changes unseen-task decisions after controlling for answer syntax.

Culinary resources cover ingredient networks, recipes, substitutions, planning, nutrition, and food knowledge (Ahn et al., 2011; Garg et al., 2018; Salvador et al., 2017; Fatemi et al., 2023; Choi et al., 2024; Wu et al., 2025; Hua et al., 2025; Eftimov et al., 2026; Jin et al., 2026). Our contribution is narrower: paired frontier-model measurement against an exhaustive culinary decision environment, rather than recipe generation or factual question answering.

9 LIMITATIONS

FlavourBench Score is agreement with a released reward map, not universal taste. Public-checkpoint rescoring and Recipe1MSubs provide convergent evidence, but do not validate the primary runtime's unrecovered training lineage, pairing and constraint rewards, sensory preference, or cooked outcomes. Within-task min–max normalization creates an interpretable 0–100 scale while discarding absolute utility magnitudes. The tasks test constrained ingredient selection, not full recipes, safety advice, embodied cooking, or long-horizon kitchen planning.

The complete-core rule removes model-specific missingness but changes the estimand to tasks every endpoint completed; roster sensitivity cannot rule out unrecorded selection effects. Results bind specific model versions, providers, routes, and collection dates. The dense maps make reward-based learning possible, but the transfer evidence comes from LoRA SFT of one 0.6B checkpoint over three seeds. It does not compare training algorithms or model scales. The public replication reuses the same trained adapters on new maps and therefore strengthens task transfer, not independent training replication.

10 CONCLUSION

FLAVOURBENCH evaluates open-ended culinary decisions without asking one language model to grade another. Exhaustive reward maps make partial credit deterministic; a rectangular common-task design makes comparisons paired; simultaneous intervals and corrected tests distinguish broad performance differences from an unsupported winner narrative. In a controlled post-training study, supervision from those maps changes decisions on unseen anchors beyond what format-matched SFT explains. The public artifacts reconstruct every reported score, contrast, training run, table, and figure.

AI USE STATEMENT

Generative AI tools assisted with methodology feedback, implementation, data cleaning and formatting, statistical-analysis support, interpretation, literature discovery, figure production, and drafting and editing the manuscript. They did not originate the empirical observations: model responses and benchmark records are retained as content-addressed artifacts, and quantitative claims are regenerated by tested analysis code. The authors reviewed the AI-assisted work; citations were checked against source records, release verifiers check the numerical artifacts, and the authors take responsibility for the final content.

ETHICS STATEMENT

No human participants were recruited. The external check uses an existing public research dataset; the release records its provenance and hashes and redistributes only aggregate validation outputs, not the raw Recipe1MSubs files. The benchmark measures a bounded culinary construct and should not be interpreted as nutrition, allergy, food-safety, or medical advice. Provider credentials and private communications are excluded from the release.

REPRODUCIBILITY STATEMENT

The anonymous supplement contains the exact task maps, model and route manifest, raw selected responses, completion records, frozen analysis plans, public-checkpoint score maps, aggregate external-validation artifact, deterministic asset builders, and an offline verifier. It also contains the reward-transfer protocol, training data, raw held-out generations, run manifests, analysis code, commands, and environment contract needed to reconstruct the leaderboard, statistical tables, and vector figures without provider access. Appendices report execution routes, protocol constants, per-family scores, and robustness analyses.

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A Complete leaderboard

Table 6: Complete leaderboard on the common 534-task core. Rank intervals use the anchor-cluster bootstrap; groups summarize the Holm-controlled pairwise graph.

Rank	Model	FB Score	simultaneous 95% CI	rank 95% CI	group
1	Grok 4.6	65.1	[61.0, 69.2]	[1, 5]	1
2	Gemini 3.1 Pro	65.0	[60.8, 69.1]	[1, 6]	1
3	GPT-5.6 Sol Pro	64.2	[60.1, 68.4]	[1, 8]	1
4	Muse Spark 1.2	63.8	[59.6, 67.9]	[1, 10]	1
5	GPT-5.6 Terra Pro	63.7	[59.5, 67.8]	[1, 11]	1
6	Claude Fable 5	63.4	[59.2, 67.5]	[2, 13]	1
7	GPT-5.6 Luna Pro	62.6	[58.5, 66.8]	[4, 14]	1
8	Claude Opus 5	62.5	[58.4, 66.6]	[3, 15]	1
9	Qwen3.8 A95B	62.1	[57.9, 66.2]	[4, 17]	1
10	Kimi K3	62.1	[57.9, 66.2]	[5, 16]	1
11	Gemini 3.6 Flash	62.0	[57.7, 66.2]	[4, 17]	1
12	DeepSeek V4 Pro 0813	62.0	[57.8, 66.1]	[5, 17]	1
13	Qwen 3.8 Max	61.5	[57.4, 65.6]	[7, 18]	1
14	Tencent HY 3	61.5	[57.3, 65.7]	[6, 19]	1
15	MiniMax M3	60.9	[56.8, 65.1]	[8, 20]	1
16	GLM 5.3	60.6	[56.4, 64.8]	[8, 20]	1
17	Muse Glimmer 30B	59.9	[55.9, 63.9]	[12, 21]	2
18	Seed 2.1 Turbo	59.7	[55.4, 64.0]	[12, 22]	2
19	Inkling	59.6	[55.4, 63.8]	[12, 22]	2
20	Claude Sonnet 5	59.5	[55.4, 63.7]	[13, 22]	2
21	GLM 5.2	58.5	[54.2, 62.7]	[16, 23]	2
22	Nemotron 3.5 Lightning	57.4	[53.2, 61.6]	[19, 25]	2
23	Command A	56.7	[52.5, 60.9]	[20, 25]	2
24	DeepSeek V4 Flash	55.4	[51.1, 59.7]	[22, 26]	2
25	Mistral Large 3	55.4	[51.4, 59.5]	[22, 26]	2
26	Llama 4 Maverick	53.7	[49.6, 57.7]	[24, 26]	3
27	Command R+	47.9	[43.7, 52.0]	[27, 27]	3

B Exact execution routes

C Task-map and protocol diagnostics

D Per-family scores

E Selection and metric sensitivity

F Public-checkpoint sensitivity

G External substitution validation

H Reward-transfer protocol and diagnostics

I Panel, task-count, pairwise, and family diagnostics

Table 7: Frozen execution routes. Automatic fallback is disabled; route changes represent complete model-block replacements, never cell pooling.

Model	Backend	Panel 1 route	Panel 2 route
Grok 4.6	openrouter	xai/zdr	xai/zdr
Gemini 3.1 Pro	openrouter	google-ai-studio	google-ai-studio
GPT-5.6 Sol Pro	openrouter	openai/flex	openai/flex
Muse Spark 1.2	openrouter	meta	meta
GPT-5.6 Terra Pro	openrouter	openai/flex	openai/flex
Claude Fable 5	openrouter	google-vertex/global	google-vertex/global
GPT-5.6 Luna Pro	openrouter	openai/flex	openai
Claude Opus 5	openrouter	azure/us	azure/us
Qwen3.8 A95B	openrouter	alibaba	alibaba
Kimi K3	openrouter	morph	morph
Gemini 3.6 Flash	openrouter	google-vertex/us	google-vertex/us
DeepSeek V4 Pro 0813	openrouter	gmicloud/fp8	gmicloud/fp8
Qwen 3.8 Max	openrouter	alibaba	alibaba
Tencent HY 3	openrouter	gmicloud/bf16	gmicloud/bf16
MiniMax M3	openrouter	venice/fp8	venice/fp8
GLM 5.3	zai_coding_direct	zai-coding-plan-direct	zai-coding-plan-direct
Muse Glimmer 30B	openrouter	fireworks	fireworks
Seed 2.1 Turbo	openrouter	seed/fp8	seed/fp8
Inkling	openrouter	deepinfra/fp8	deepinfra/fp8
Claude Sonnet 5	openrouter	amazon-bedrock/claude-on-aws	amazon-bedrock/claude-on-aws
GLM 5.2	openrouter	coreweave/fp4	coreweave/fp4
Nemotron 3.5 Lightning	openrouter	venice/fp4	venice/fp4
Command A	openrouter	cohere	cohere
DeepSeek V4 Flash	openrouter	deepinfra/fp4	deepinfra/fp8
Mistral Large 3	openrouter	mistral	mistral
Llama 4 Maverick	openrouter	digitalocean	digitalocean
Command R+	openrouter	cohere	cohere

Table 8: Frozen task-map diagnostics. Chance is the uniform mean over all 56 portfolios; top gap is the median best–second-best score difference.

Family	Exact chance	Top gap	Distinct scores
Substitution	45.2	5.6	56
Pairing	45.0	5.6	56
Constraints	3.4	37.3	4

Table 9: Primary design and inference constants.

Quantity	Frozen value
Models	27
Panels / families	2 / 3
Tasks per model	534
Unique anchor clusters	534
Portfolios per task	56
Complete cells	14418
Bootstrap replicates	50000
Sign-flip replicates	100000
Pairwise hypotheses	351
Common-core rule	status/parser only; score-blind

Table 10: Mean score by family. Every row uses identical tasks.

Model	Substitution	Pairing	Constraints
Grok 4.6	66.1	70.9	58.2
Gemini 3.1 Pro	69.0	71.6	54.2
GPT-5.6 Sol Pro	66.2	72.5	54.1
Muse Spark 1.2	65.9	70.3	55.0
GPT-5.6 Terra Pro	66.1	72.1	52.8
Claude Fable 5	68.2	70.6	51.3
GPT-5.6 Luna Pro	64.5	69.7	53.7
Claude Opus 5	67.2	69.9	50.4
Qwen3.8 A95B	66.0	67.8	52.5
Kimi K3	65.5	73.7	47.0
Gemini 3.6 Flash	68.1	66.2	51.6
DeepSeek V4 Pro 0813	66.4	69.5	49.9
Qwen 3.8 Max	66.6	68.5	49.5
Tencent HY 3	66.0	67.9	50.5
MiniMax M3	65.4	69.9	47.6
GLM 5.3	63.7	69.6	48.4
Muse Glimmer 30B	65.1	66.8	47.8
Seed 2.1 Turbo	65.3	66.2	47.6
Inkling	63.6	65.8	49.5
Claude Sonnet 5	65.3	69.2	44.1
GLM 5.2	63.6	67.2	44.5
Nemotron 3.5 Lightning	65.3	64.3	42.5
Command A	64.7	63.9	41.6
DeepSeek V4 Flash	63.1	63.2	40.1
Mistral Large 3	60.4	65.8	40.0
Llama 4 Maverick	59.4	61.4	40.2
Command R+	56.7	54.9	32.0

Table 11: Alternative summaries of the same 534 decisions.

Score summary	Rank ρ	Pair order	Point leader
FlavourBench Score	1.000	100.0%	Grok 4.6
Chance-adjusted gain	0.985	96.0%	Gemini 3.1 Pro Preview
Action percentile	0.969	93.4%	Grok 4.6
Exact-optimum rate	0.938	89.2%	Grok 4.6

Table 12: Aggregate agreement with the primary reward map.

Reward map	Task-map ρ	Model-rank ρ	Pair order	Point leader
Epicure-Cooc	0.752	0.957	91.7%	Grok 4.6
Epicure-Core	0.672	0.915	88.0%	Grok 4.6
Epicure-Chem	0.660	0.903	86.9%	Grok 4.6

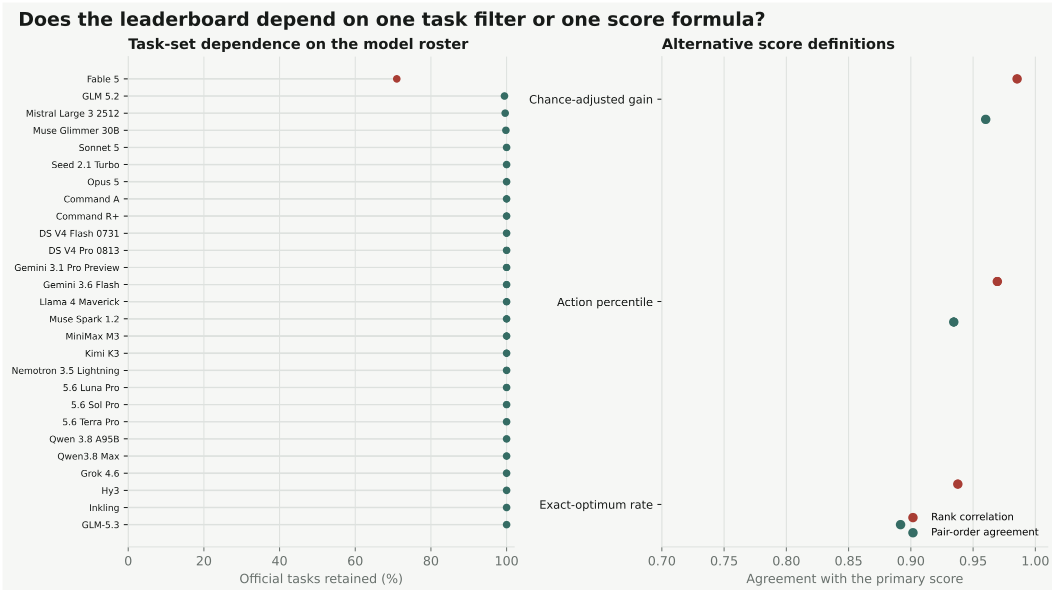

Figure 7: Left: task overlap after omitting each endpoint and rerunning frozen selection. Right: agreement between the primary metric and alternative summaries.

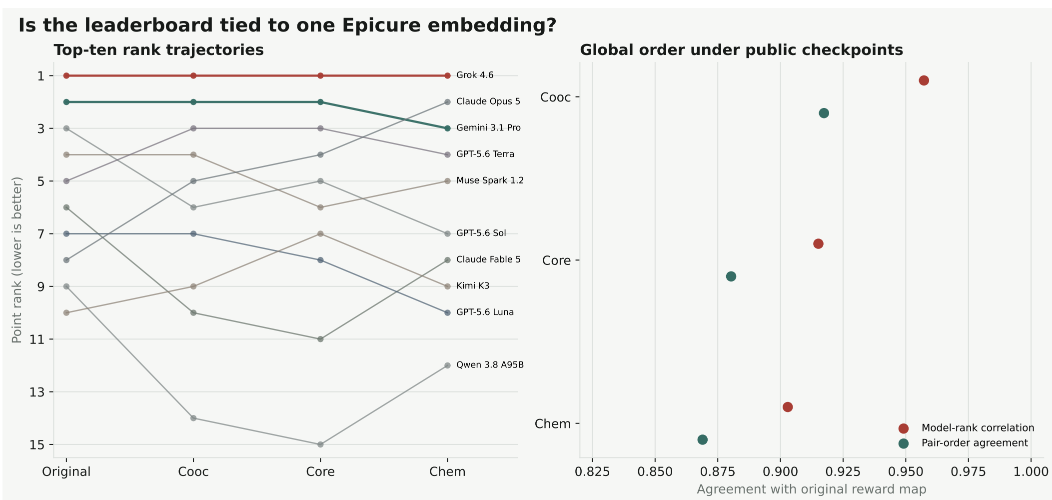

Figure 8: Rank trajectories and aggregate agreement after replacing the reward map while holding all tasks and model selections fixed.

Table 13: Frozen reward-transfer protocol. Validation loss is monitored but never selects a checkpoint.

Component	Frozen value
Base	Qwen3-0.6B, revision `c1899de289a04d1...`
Conditions	unmodified base; format-control SFT; Epicure-optimum SFT
Train / validation / transfer	270 / 72 / 84 anchor-disjoint tasks
Training seeds	20260824, 20260825, 20260826
LoRA	rank 16; alpha 32; dropout .05; all linear layers
Optimization	3 epochs; AdamW; LR 10^{-4}; effective batch 16
Checkpoint rule	final adapter only; no validation selection
Decoding	greedy; thinking disabled; 64 new tokens
Primary score	equal family and panel; unparseable responses score zero
Inference	50,000 crossed bootstraps; 100,000 anchor sign flips

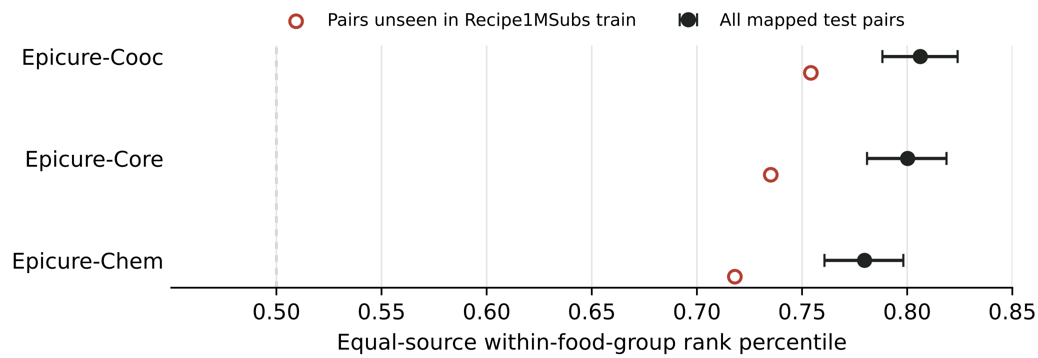

Figure 9: Observed Recipe1MSubs targets ranked against same-food-group alternatives. Filled points use all mapped test pairs; open points retain only pairs absent from the training split.

Table 14: Descriptive reward-transfer scores by task family. Δ is Epicure SFT minus format control; these family rows were not separate confirmatory hypotheses.

Evaluation	Family	Base	Format control	Epicure SFT	Δ
Transfer	Substitution	57.93	41.62	52.79	+11.16
Transfer	Pairing	28.66	46.34	53.56	+7.23
Transfer	Constraint	3.39	4.73	26.24	+21.51
Public	Substitution	47.63	47.43	52.39	+4.96
Public	Pairing	34.36	43.59	53.62	+10.02
Public	Constraint	3.69	3.78	23.99	+20.21

Table 15: Matched training-seed scores. Each Δ compares adapters initialized with the same seed.

	Anchor-disjoint transfer			Public-map replication		
Seed	Control	Epicure	Δ	Control	Epicure	Δ
20260824	31.61	44.47	+12.86	32.06	44.74	+12.69
20260825	31.19	44.53	+13.34	31.10	42.40	+11.30
20260826	29.89	43.59	+13.70	31.65	42.85	+11.20

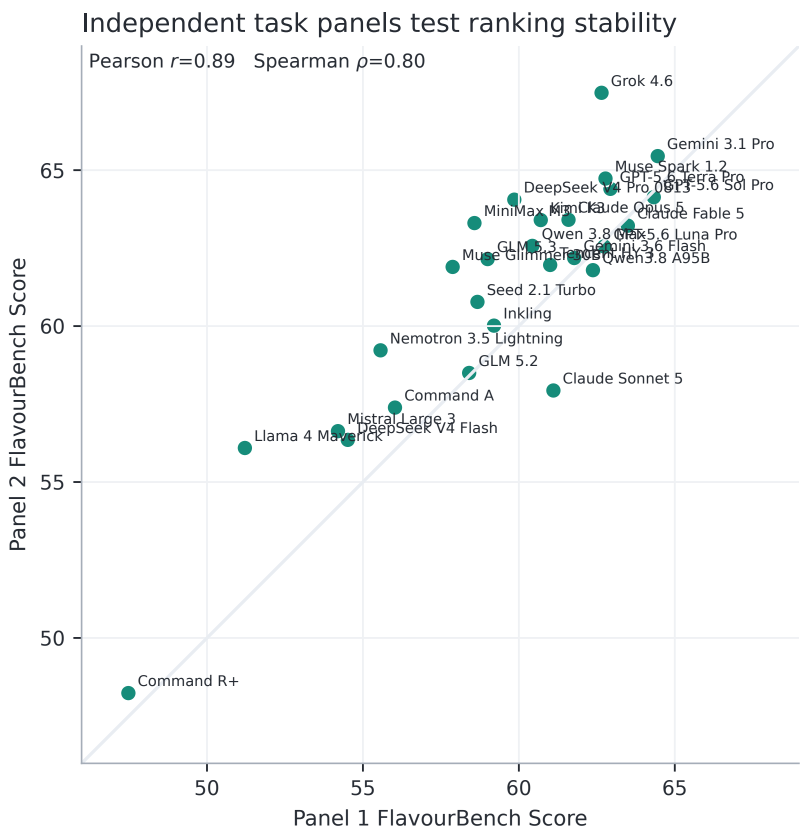

Figure 10: Independent-panel scores for all 27 endpoints.

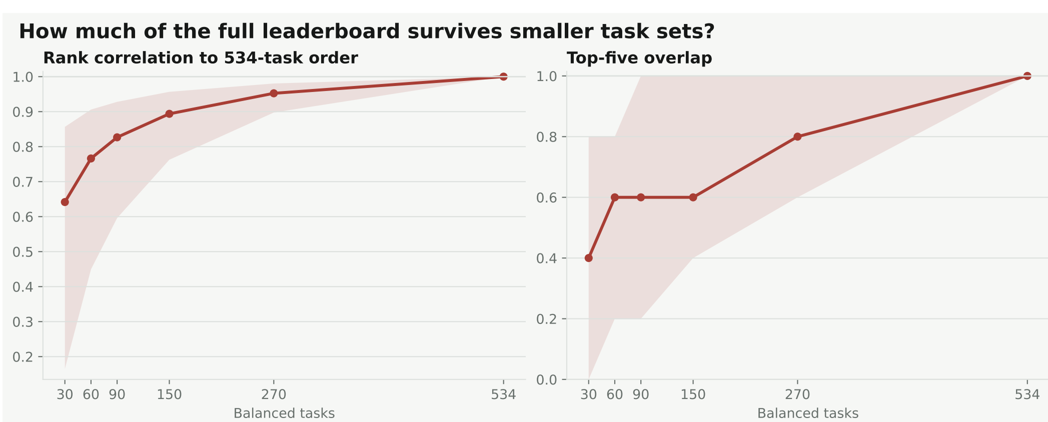

Figure 11: Stability of score-blind task subsets relative to the complete point order.

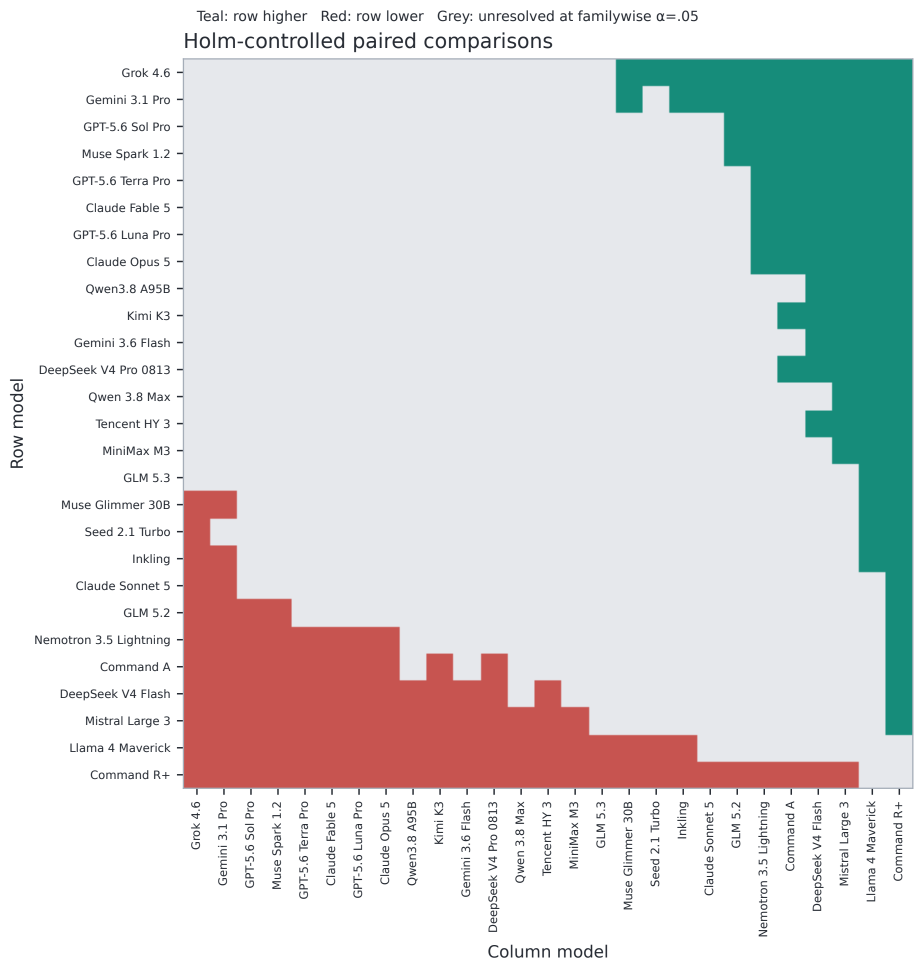

Figure 12: All paired endpoint contrasts after Holm correction. Grey cells are unresolved.

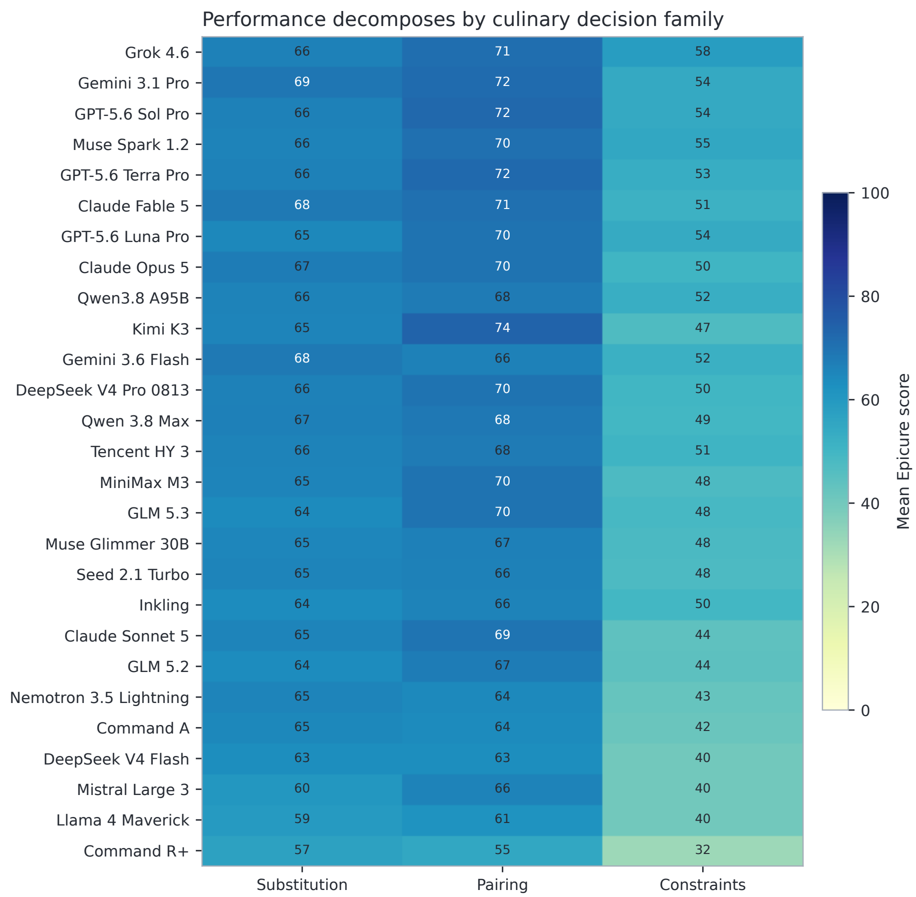

Figure 13: Mean score by culinary decision family. Each column contains 178 identical tasks per endpoint; the primary score gives the three families equal weight.

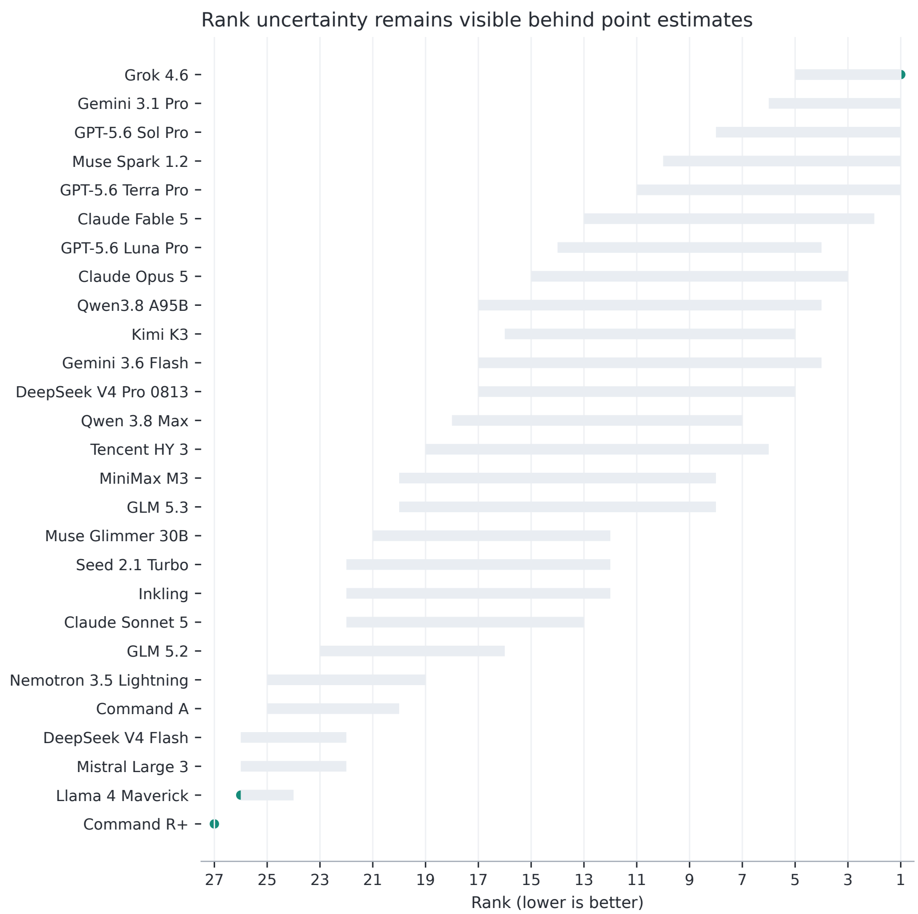

Figure 14: Point ranks and 95% bootstrap rank intervals.